

SVM (Support Vector Machines) ML Algorithms

Tasks

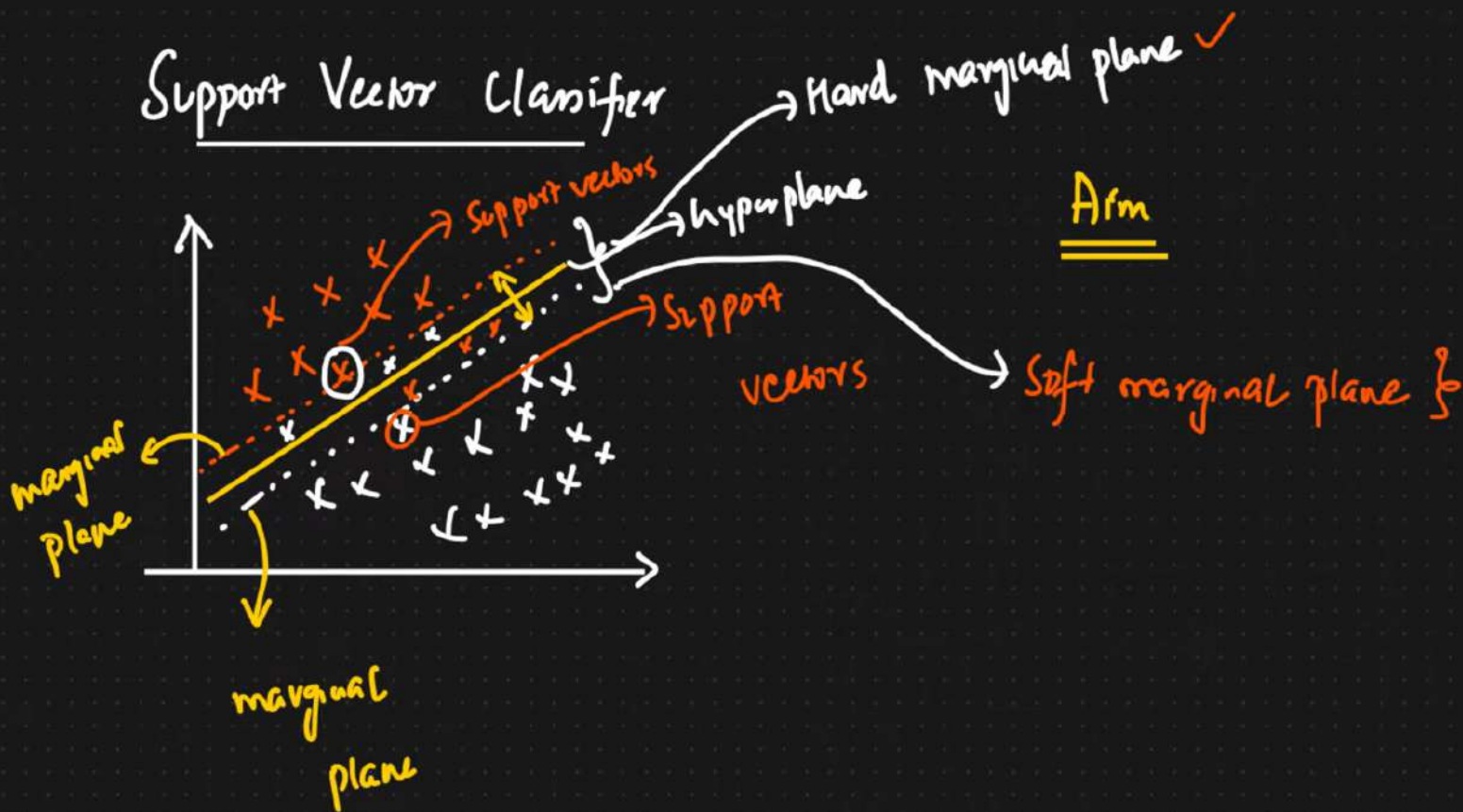
- ① Classification ✓
- ② Regression

Support Vector Classifier

① Classification

② Regression

Support Vector Classifier



$$by = -ax - c$$

$$y = -\frac{a}{b}x - \frac{c}{b}$$

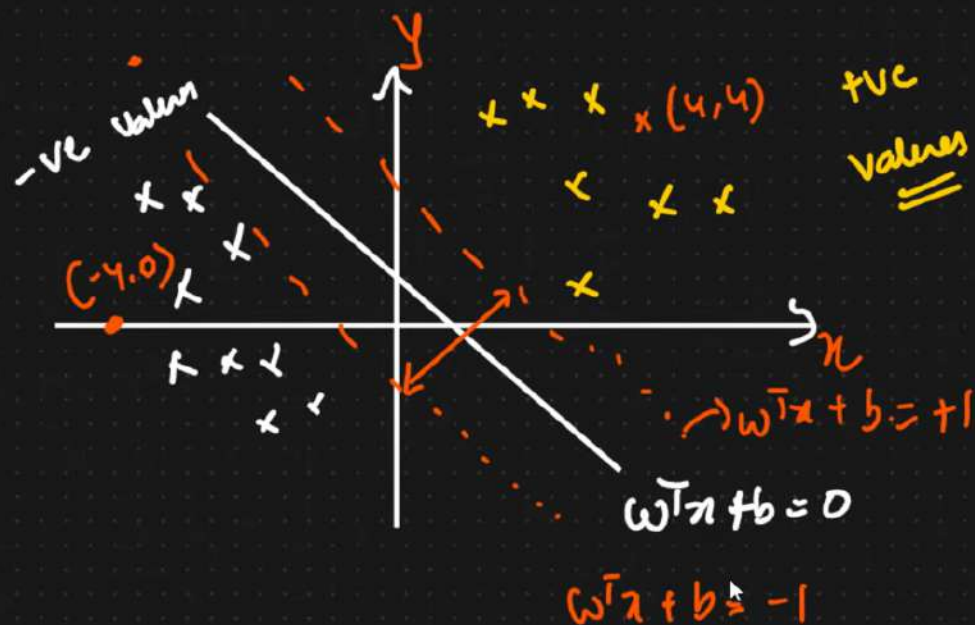
$$m = \frac{-a}{b}$$

$$c = \frac{-c}{b}$$

$$3(4) + 2(4) + 4 = 0$$

$$12 + 8 + 4$$

$$\boxed{24} \Rightarrow +ve$$



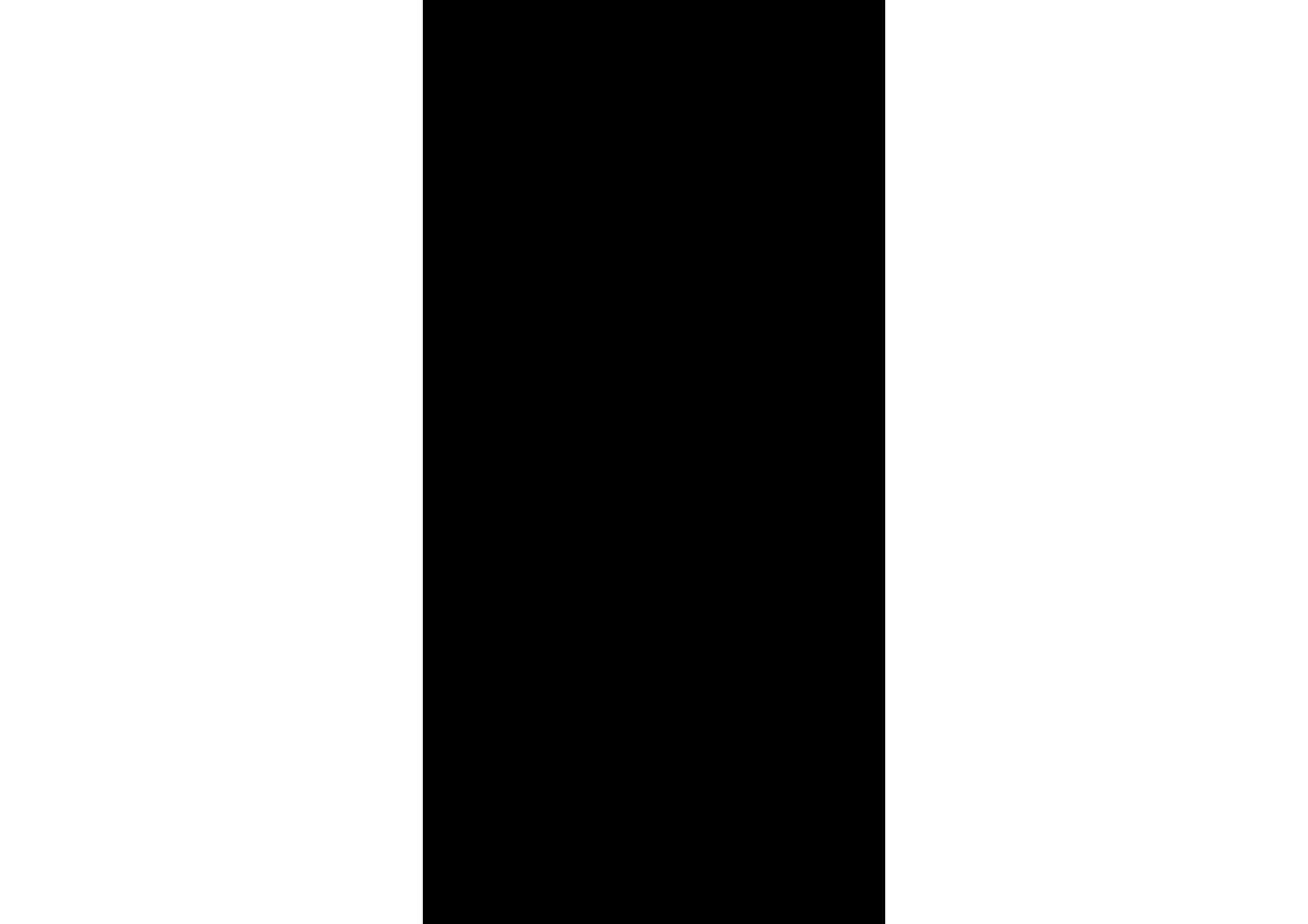
$$3x + 2y + 4 = 0$$

$$3(-4) + 2(0) + 4$$

$$-12 + 0 + 4$$

$$= \boxed{-8}$$





$$w^T x_1 + b = +1$$

$$w^T x_2 + b = -1$$

$$\begin{array}{ccc} (-) & 2 & (-) \quad (+) \end{array}$$

$\vec{w}$

$$\left. \frac{w^T (x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|} \right\} \Rightarrow \text{Margin}$$

(w, b)

$\|w\|$

Constraint

Show that

$$y_i \left\{ \begin{array}{ll} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{array} \right\}$$

$$\boxed{y * (w^T x_i + b) \geq 1} \Rightarrow \text{constraint}$$

$$\text{Maximize}_{(w,b)} = \frac{2}{\|w\|}$$

$$\Rightarrow \boxed{\text{Minimize}_{(w,b)} = \frac{\|w\|}{2}}$$

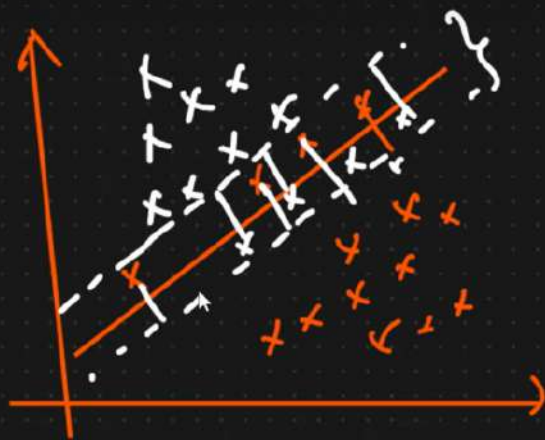


Cost function : Minimize $\frac{||w||}{2} + C_i \sum_{i=1}^n \xi_i$

$\{ \text{Summation of the distance of misclassification points from margin plane} \}$

$\{ \text{How many points we can} \}$

$C=8$



Avoid misclassification}



23:56 / 25:24



More videos



YouTube